

Test: Observability, Observers, and Observer Based Control

1. Is the following system controllable?

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} -3 & 0 \\ 1 & -2 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t) \\ y(t) &= \begin{bmatrix} 0 & 1 \end{bmatrix} x(t)\end{aligned}$$

Controllability matrix

$$C = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & -2 \end{bmatrix}$$

The rank of C is one, i.e., not two ($\neq n$). Thus, the system is not controllable.

2. Is the following system observable?

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} -3 & 0 \\ 1 & -2 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t) \\ y(t) &= \begin{bmatrix} 0 & 1 \end{bmatrix} x(t)\end{aligned}$$

Observability matrix

$$O = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$$

$\det(O) = -1$, i.e. the rank of O is two. Thus, the system is observable.

3. Consider the one-dimensional system

$$\begin{aligned}\dot{x}(t) &= -x(t) + u(t) \\ y(t) &= x(t)\end{aligned}$$

Design an observer gain L such that the observer has an eigenvalue at -5 .

Observer eigenvalues are

$\text{eig}(A+LC)$, where $A+LC = -1 + L \cdot 1$. Therefore, $L = -4$.

4. Consider the following system

$$\dot{x}(t) = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(t)$$

with state feedback

$$u(t) = \begin{bmatrix} -2 & -2 \end{bmatrix} x(t).$$

What are the eigenvalues of the closed-loop system?

The closed-loop system has eigenvalues

$$\begin{aligned}\text{eig}(A+BF), \text{ where } A+BF &= \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} -2 & -2 \end{bmatrix} \\ &= \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}\end{aligned}$$

Therefore, the closed-loop poles are both in -1 .